(ii) Is ρ symmetric? If not, state the reason, write minimum number of ordered pairs to be included to ρ so as to make it symmetric and write minimum number of ordered pairs to be deleted from ρ so as to make it symmetric.
(iii) Is ρ transitive? If not, state the reason, write minimum number of ordered pairs to be included to ρ so as to make it transitive and write minimum number of ordered pairs to be deleted from ρ so as to make it transitive.
(iv) Is ρ an equivalence relation? If not, write the minimum ordered pairs to be included to ρ so as to make it an equivalence relation.

Solution:

(i) ρ is not reflexive because $(3, 3)$ is not in ρ. As $(1, 1)$ and $(2, 2)$ are in ρ, it is enough to include the pair $(3, 3)$ to ρ so as to make it reflexive.
(ii) ρ is not symmetric because $(1, 2)$ is in ρ, but $(2, 1)$ is not in ρ. It is enough to include the pair $(2, 1)$ to ρ so as to make it symmetric.
It is enough to remove the pair $(1, 2)$ from ρ so as to make it symmetric
(iii) ρ is not transitive because $(3, 1)$ and $(1, 3)$ are in ρ, but $(3, 3)$ is not in ρ. To make it transitive we have to include $(3, 3)$ in ρ. Even after including $(3, 3)$, the relation is not transitive because $(3, 1)$ and $(1, 2)$ are in ρ, but $(3, 2)$ is not in ρ. To make it transitive we have to include $(3, 2)$ also in ρ. Now it becomes transitive. So $(3, 3)$ and $(3, 2)$ are to be included so as to make ρ transitive.
But if we remove $(3, 1)$ from ρ, then it becomes transitive.
(iv) We have seen that

- to make ρ reflexive, we have to include $(3, 3)$;
- to make ρ symmetric, we have to include $(2, 1)$;
- and to make ρ transitive, we have to include $(3, 3)$ and $(3, 2)$.

To make ρ as an equivalence relation we have to include all these pairs. So after including the pairs the relation becomes $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1), (3, 2)\}$

But this relation is not symmetric because $(3, 2)$ is in the relation and $(2, 3)$ is not in the relation. So we have to include $(2, 3)$ also. Now the new relation becomes

$$\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1), (3, 2), (2, 3)\}$$

It can be seen that this relation is reflexive, symmetric and transitive, and hence it is an equivalence relation. Thus we have to include $(3, 3), (2, 1), (3, 2)$ and $(2, 3)$ to ρ so as to make it an equivalence relation.

Now let us learn how to create relations having certain properties through the following example.

Example 1.12 Let $A = \{0, 1, 2, 3\}$. Construct relations on A of the following types:

(i) not reflexive, not symmetric, not transitive.
(ii) not reflexive, not symmetric, transitive.
(iii) not reflexive, symmetric, not transitive.
(iv) not reflexive, symmetric, transitive.
(v) reflexive, not symmetric, not transitive.
(vi) reflexive, not symmetric, transitive.
(vii) reflexive, symmetric, not transitive.
(viii) reflexive, symmetric, transitive.